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Comparative review and evaluation of state-of-the-art photovoltaic cooling technologies

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ABSTRACT

Photovoltaic systems have a reliable method to generate electrical power from direct solar irradiance. Their efficiency is only up to 20% in real operating circumstances and is affected adversely by the inevitable temperature rise in their surface. Passive, active or combined cooling methods are applied to reduce operating temperature to a suitable level, depending on specific application. A Great body of literature has been investigating different PV cooling methods and techniques to reach optimized system efficiency and cooling effect. It is critical for both researchers and practitioners to have a thorough understanding and awareness of the most recent advances and existing gaps. Although there are certain review papers in this field, it is necessary to have up-to-date reviews of the most recent progress and research since the previous ones. In the present work, the main focus was to provide a review of the most recent studies and to compare their outcomes to provide a reference for practitioners and researchers. The holistic classification and evaluation of all the possible cooling technologies were provided in order to draw a comparison between various cooling approaches. Based on the analysis obtained, cooling techniques can ensure improvement in electrical efficiency in a range from 1% to 53% with expected decrease in the photovoltaic panel's surface operating temperature from 1.8 °C to 46 °C. It was also found that spectrum filtering and jet impingement indicated highly remarkable enhancements. The other factors, such as the size of the examined photovoltaic panel, photovoltaic technology, as well as climate circumstances have a strong impact on the cooling performance besides the selection of the specific coolant. The necessity for integral evaluation (performance, economic and environmental) of cooling techniques was also identified as key issue.

1. Introduction

Meeting cities' energy demands in the era of climate change is one of the most critical challenges of our time. As a result of climate change and global warming, energy consumption has increased (Santamouris et al., 2001, 2015) and on the other hand, conventional energy supply contributes directly to the current trend of global warming and greenhouse gas emissions (Edenhofer et al., 2011). Renewable energy resources are recognized to be the most dependable options to address this problem, such as solar energy. Solar energy is an omnipresent and endless source of electrical and thermal power. Photovoltaic systems are the most efficient technologies for extracting electrical energy from direct solar irradiance and have gained increasing popularity over the last decades (Dwyer and TeskeRenewables, 2018; Grubišić-Čabo et al., 2016). They account for more than half (60%) of the world's renewable energy resource and are estimated to surge up by 2026 (IEA and IEA, 2021). Fig. 1 clearly indicates the great proportion of solar PV systems in the renewables and the potential growth of them soon to be the leading renewable power of the future.

However, the energy conversion rate of the available commercial PV technologies usually ranges between 5 and 20%. Unfortunately, this rate decreases as a result of constant exposure of the PV panel to a high operating temperature. Usual efficiency degradation ranges from 0.25% to 0.5% per degree Celsius (Grubišić-Čabo et al., 2016; Hussein and Aboaltabooq, 2021; Kalaiselvan et al., 2018), depending on the type of PV technology used. That is to say, the overheating of the PV panel also

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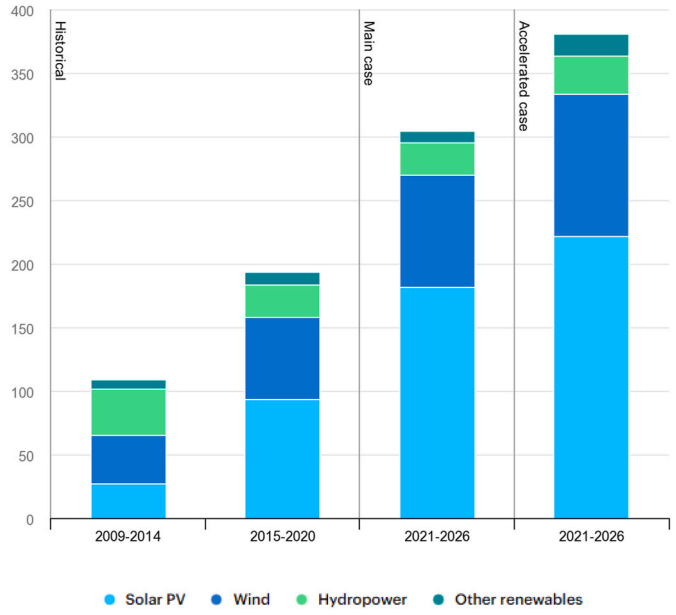


Fig. 1. Technology-driven growth in renewable electricity capacity, main and accelerated cases, 2015–2020 and 2021–2026 (IEA and IEA, 2021).

has adverse effect on the lifetime of the whole PV system (Perrakis et al., 2020), and so it has economic loss as well.

To ensure efficient performance of the PV panel, the operational temperature of the PV panel needs to remain as low as possible, or at Standard Test Condition (STC) which is 25 °C temperature (Kalaiselvan et al., 2018). Therefore, cooling down the PV systems while sustaining their efficiency through effective methods is a growing challenge and concern. A considerable body of research and experiments have been done to find the optimum methods, techniques and technologies. There are also cases where PV panel is incorporated within the multi energy-generation systems, such as photovoltaic-thermal systems (PVT), where temperature level of the PV panel should be carefully balanced through proper design for specific climate conditions, to ensure reasonable electrical and thermal performance. These methods generally could be classified into three classes of passive, active and combined cooling approaches, based on their energy consumption. Key factors needed to be taken into consideration adopting appropriate cooling technique for the PV system are uniform distribution of the cooling on the panel's surface, minimum power consumption, reliability, cost-effective installation and maintenance, weather and climatic conditions, etc. (Grubišić-Čabo et al., 2016; Royne et al., 2005). Climate condition is a key factor to consider proper design for certain cooling approach.

In active cooling method, an external equipment or device which consumes operational energy is attached to the panel in the forms of fans, sprays, pumps, etc. (Kalaiselvan et al., 2018), to ensure forced and constant circulation of the coolant (Hussein and Aboaltabooq, 2021), which could be water, air or Nano fluids. These methods are usually more efficient in results, specifically in hot regions (Gharzi et al., 2020), but the power consumption to production ratio is the main factor that justifies techno-economic aspect of these techniques (Hussein and Aboaltabooq, 2021; Gharzi et al., 2020; Dwivedi et al., 2020). Moreover, due to their difficulties and complexities in installation and maintenance, some cooling strategies, besides expensive costs (Li et al., 2021a), are primarily impracticable and less used in the real world and domestic uses (Li et al., 2021a; Chehreh, 2021). Unlike active methods, passive cooling refers to approaches that remove heat from the PV cells with no additional energy consumption (Kalaiselvan et al., 2018). They are relatively easier to install, and in some cases more economical (Dwivedi et al., 2020).

The main aim of this study is holistic classification of all the possible photovoltaic cooling technologies. The novelty of the present study is that unlike former reviews, a comparative approach was adopted to firstly evaluate and assess different research and methods of a specific cooling technique. And ultimately to compare and investigate all techniques together in order to frame and highlight their advantageous and disadvantageous all together.

2. Review methodology and necessity for conducted review

In this review work, the focus was to present a holistic classification and evaluation of available cooling technologies as well as to investigate the most recent studies in the field. In order to do so, at first, the recent review papers were examined to identify existing gaps and to present advancements in the field. The results of the investigation conducted are presented in Table 1. Next, after extracting a classification for all the possible PV cooling techniques, each category was searched with its key words in google scholar in the time frame limited to 2011, to cover a decade. Relevant papers and research were selected based on some

Table 1
Highlights of investigated review studies on PV cooling techniques.

Review paper	Research highlights
Chehreh (Chehreh, 2021), 2021	Technical detail and classification of cooling methods, weakness/strength
Kandeal, Algazzar (Kandeal et al., 2021), 2021	Review of Nano-based cooing systems (evaluation and performance)- comprehensive bibliometric analysis and the research trend using VOS viewer software based on the Scopus database
Salameh, Zhang (Salameh et al., 2021), 2021	The latest advances in solar PV technologies across all generations + Life cycle assessment analysis (LCAA) including environment and economy, countries statistics, Comparison of cooling methods based on electrical and thermal efficiency
Gharzi, Arabhosseini (Gharzi et al., 2020), 2020	environmental condition and host country of research, advantageous and disadvantageous, focus of studies, thermal aspects, electrical efficiency enhancement, technology type used, and cooling effect were discussed
Zhang, Shen (Zhang et al., 2020), 2020	Technical detail and classification of cooling methods, with environmental and design factors, comprehensive comparison of electrical, thermal and cooling efficiency
Dwivedi, Sudhakar (Dwivedi et al., 2020), 2020	Environmental parameters, investigating different cooling methods advantageous and disadvantageous (not unified), challenges and future outline
Sato and Yamada (Sato and Yamada, 2019), 2019	state-of-the-art technologies, a summary of the cooling effects and methodologies of various cooling methods, radiative cooling method is quantitatively evaluated
Alsayah, Aboaltabooq (Alsayah et al., 2019a), 2019	review of the methods with a focus on the type of fluid that cools the cell, costs of electricity consumption and their economic feasibility, difference between each, advantageous and disadvantageous
Siecker, Kusakana (Siecker et al., 2017), 2019	classified based on their focus (focusing on arrays for reducing surface temperatures), advantages and disadvantages, area of application, contribution and the type of technology used, techno-economic character, comparing the performance of each
Chandel and Agarwal (Chandel and Agarwal, 2017), 2017	Seeking reliable performance and commercially viability of PCMs, considering geographical location
Zubeer, Mohammed (Zubeer et al., 2017), 2017	focus on small PV panels for domestic use, advantages and disadvantages of some techniques
Sainthiya and Beniwal (Sainthiya and Beniwal, 2017), 2017	An examination of the utility and importance of optimization techniques for different types of cooling systems

factors. Firstly, numerical and comparable data regarding temperature decrease and electrical efficiency were examined. Therefore, those papers that did not present a clear result to compare were excluded (usually it was due to lack of sufficient technical data to provide comparison). Secondly, on each category, the focus was on the most recent and key studies with novelty in methods and techniques with a clear methodology. In the end, the selected papers were investigated based on comparable criteria in a table on each category, which defines the limitations of the study. On this basis, the reviews were limited on analysis of specific parameters namely, PV technology, cooling configuration and method, the coolant or material properties, performance improvement, temperature decrease, climate condition and location of the study.

The process of conducted research and gained results were also summarized. Based on our studies and to the best of our knowledge, there was no recent review paper which has had presented overview and evaluation of cooling including all the PV cooling techniques and technologies. Therefore, this paper has covered all the existing cooling techniques and classified them based on their methods. The electrical efficiency improvements and temperature decrease regarding the climate and the PV cell technology, were selected as the main parameters for comparison.

3. Types of cooling techniques

A well-known classification of cooling systems is based on their energy consumption. Active cooling approaches requires extra energy for cooling by using for instance pumps or fans, etc., for heat extraction, while passive cooling systems don't require operational energy. A coolant (usually fluid) is needed in specific cooling approach such as water, air, nanofluids, etc. Active cooling systems' performance is more efficient, but usually more expensive due to the needed operational energy (Kalaiselvan et al., 2018) and more complex design in general. However, in the case of the active cooling techniques the performance benefit should justify overall investment. The cooling techniques for PVs have been classified based on their cooling methods, used technologies and cooling medium, Fig. 2. However, environmental conditions (wind speed, annual radiation, temperature, humidity, etc.) should be the determinative factor for choosing the appropriate system and technique (Chehreh, 2021) in order to secure economic viability.

3.1. Passive cooling

Through the passive cooling mechanism, the extra heat is absorbed and dissipated by convection or natural circulation of the coolant to the surrounding environment with no request for operational energy. As this technique is applied to the back surface of the PV panels, the size of the PV panel, strongly affects the performance of the considered passive cooling approach (Kalaiselvan et al., 2018) Although passive cooling techniques are usually economical, cost-effective, and easily applied, they are generally weak in performance and slow in cooling achievement (Kalaiselvan et al., 2018; Gharzi et al., 2020; Dwivedi et al., 2020). Nonetheless, combining it with hybrid thermal technologies known as PVT system can offset its drawbacks with the combination of thermal and electrical efficiency to generate heat for domestic use and take advantage of the increased energy output per unit (Chehreh, 2021; El-Seesy et al., 2012). A wide variety of materials may be used, ranging from metals or fluid medium systems with high thermal conductivity to phase change materials (PCM) or heat pipes with more complex structures. The specific passive cooling approaches will be elaborated in the continuation and followed with discussion.

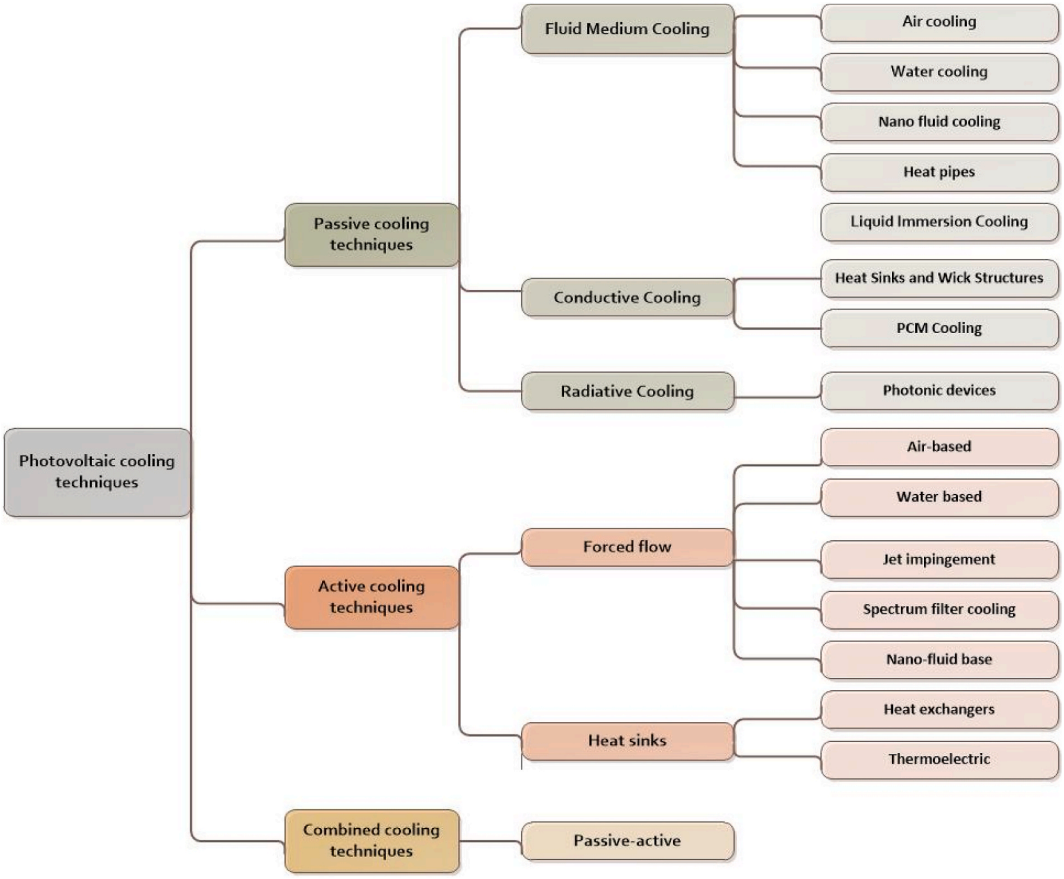


Fig. 2. Classification of PV cooling techniques.

3.1.1. Fluid medium cooling

Using Fluid coolants to decrease the temperature of the PV systems could be implemented on the backside or front side of the PV panels, which have been widely used (Zhang et al., 2020; Bijjargi et al., 2016). The selection of the working fluid determines thermal properties and that are key ones to enable more efficient cooling, as well as uniformity of the cooling and which is also an important aspect. In general, the air-based cooling would secure less performance improvement when compared to the water-based cooling, while on the other side the water is valuable resource in specific geographical areas and also evaporates. In that sense, the water-based cooling should be justified. When considering existing research studies, the selection of the working fluid is influential for cooling efficiency as well as general flow circumstances that are affecting heat transfer rate. Most of the studies have been considering mostly air or water as coolant. The water is more appropriate as coolant when compared to the air, however water evaporates (specific cooling setups) and on some locations water is limited resource. The use of air is usually linked with simpler designs of cooling approaches with lower expected cooling performance. However, use of air as coolant is beneficial for specific configurations, especially building integrated photovoltaics. Schematic diagram of the following passive fluid medium cooling approaches can be found in Fig. 3.

3.1.1.1. Passive air cooling. Through passive air cooling, with no additional material and in a simple way, the air naturally flows on the front or back surface (the back surface is more efficacious (Dwivedi et al., 2020)) of the PV panel. It uses natural or forced, convection, i.e. depending from the operating circumstances and cooling setup (free--standing, building integrated, etc.), to control the overheating of the modules. Due to its simplicity, environment friendly, marginal volumetric change at low temperature, and small cost, it is actually the most popular and used method in real applications, especially in severely cold climates (Dwivedi et al., 2020; Good et al., 2015; Kamel and Fung, 2014). With respect to the previously mentioned facts, the proper design of passive cooling is key one in order to secure reasonable performance improvement and by that economic suitability. Mittelman, Alshare (Mittelman et al., 2009) have proposed air passive cooling as an appropriate method for rooftop integrated PV panel by contemplating open channels between PV panels and the roof. Based on numerical analysis and scale assessment, they have proposed channel aspect ratios between 15 and 50 and inclination angles between 30° and 90°. The study was obtained only as numerical investigation, however the experimental verification and further investigation in that direction is necessary in order to get credible data regarding the performance improvement.

In recent years, researchers have adopted new innovative methods in passive air cooling, which are reviewed in the following (Table 2). Introducing a new technique by using the natural vapor as heat transfer fluid in their study, Ebrahimi, Rahimi (Ebrahimi et al., 2015) have suggested the establishment of panel on river canals and places in which vaporization exists naturally. Utilizing the impacts of the natural vapor flow rate, natural vapor temperature and natural vapor distribution for cooling performance, they have observed a significant drop in PV cell temperature in their study. The optimum performance was also suggested at a high natural vapor flow rate and low natural vapor temperature. However, the critical issue in these moist environments could be durability of the PV systems in these environments and which is an aspect that should be carefully investigated. Moreover, an environmental impact should be also investigated.

Zou, Gong (Zou et al., 2017) have run CFD simulations and numerical analysis to assess a novel passive cooling equipment on High Concentration Photovoltaic (HCPV) systems called Solar Enhanced Passive Air Cooling System (SEPACS), which, unlike the conventional passive air-cooling techniques, does not rely on the ambient temperature and the wind; instead, by harnessing solar irradiance, it can automatically create an airflow that enhances convective heat transferring between air and PV cells. Adopting a novel method which contributes a sorption-based atmospheric water harvester as a cooling device Li, Shi (Li et al., 2020), decreased the cell's temperature by at least 10 °C, while increasing the electrical generation by 13%–19%. In this system, the atmospheric water vapor is collected by an atmospheric water harvester (AWH), which is directly attached to the back of the solar panel absorbs heat from the panel throughout the day and releases it throughout the night. This approach was tested both in summer and winter and has the potential for future developments. The study didn't consider an economic evaluation, and which is necessary to understand overall viability of the cooling approach.

3.1.1.2. Passive water cooling. Using water as a coolant is considered considerably efficient and clean regarding its high thermal capacity and being an environmentally friendly fluid with lower risks of leaking toxic chemicals to the surroundings. (Kalaiselvan et al., 2018; Shahverdian et al., 2021a, 2021b). However, the interaction of the water with PV panel should be carefully investigated with respect to the environmental constraints, since it is not fully clear what are the long-term environmental impact associated with fact that water is in the close contact with PV panel. In comparison to passive air cooling, water cooling is more advantageous and efficient, regarding its specific heat capacity and higher thermal conductivity (Zhang et al., 2020; Hamid et al., 2014) as already highlighted. Unlike active water-cooling techniques in which water is circulated by a pump or other components on the panel, in passive cooling approach, the water is not circulating by forced way, therefore, cooling systems are much simpler in terms of design (Chehreh, 2021; Li et al., 2020), and there is no request for operational energy to manage cooling.

The physical properties and parameters, the inlet water temperature, as well as the ambient temperature have a significant impact on the efficiency of water-cooling systems for PVs (Bigorajski and Chwieduk, 2019; Wu et al., 2018a; Croitoru and Badea, 2013). Other relevant environmental factors are linked with specific geographical location where for instance solar radiation is one of key factors, as the more intense radiation is, the more intense temperature increase would occur. The wind speed is also important factor that affects heart transfer circumstances, i.e., cooling performance. Namely, for 1.0nullm/s rise in the wind speed leads to about 3.3 °C drop in cells' temperature (Bigorajski and Chwieduk, 2019; Wu et al., 2018a). They mostly take



(a) Air cooling (b) Evaporative (c) Heat pipes (d) Liquid immersion

Fig. 3. Schematic diagram of passive fluid medium cooling techniques (Gharzi et al., 2020; Sato and Yamada, 2019).

Table 2
Highlights of photovoltaic system passive air-cooling studies.

Research	Year	PV cells	System	Technique	Temperature decrease	Electrical efficiency gain	Climate	Location
Li, Shi (Li et al., 2020)	2020	NG	PV	atmospheric water sorption–evaporation cycle	>10 °C	13–19%	_	Saudi Arabia
Zou, Gong (Zou et al., 2017)	2017	Monocrystalline	HCPV	Solar Enhanced Passive Air Cooling System (SEPACS)	Kept it below 75 °C	NG	_	_
Ebrahimi, Rahimi (Ebrahimi et al., 2015)	2015	NG	PV	natural vaporization	6–7 °C	12.12–22.9%	_	Iran
Mittelman, Alshare (Mittelman et al., 2009)	2009	NG	PV	Open air channels	10–20 °C	1–2%	_	_

advantage of the evaporative effect, using the latent heat capacity of the water to expel considerable amounts of heat (Gharzi et al., 2020). Previously mentioned technique could be implemented by different techniques to transport the water using capillary effects such as clay pot, cotton wick, porous foils coating, etc., for better evaporation.

According to Belyamin, Fulazzaky (Belyamin et al., 2021), based on experimental methods and linear models, the electrical efficiency of the solar panels and the characteristics of the solar panel power curve is directly affected by the flow rate of cooling water. Moreover, they have found that around noontime, thermal and electrical power, as well as power loss reach their maximum value in PVT panel cooled by passive water flow. The examined PVT setup was not innovative one since this kind of the design was already previously investigated in other research studies.

El-Seesy, Khalil (El-Seesy et al., 2012) In another research, Chandrasekar, Suresh (Chandrasekar et al., 2013) used cotton wick structure with water wrapped spirally at the back of the panel, and immersed in Al2O3/water nano-fluid and CuO/water nano-fluid. In this technique, the excess heat is absorbed based on capillary effect. Chandrasekar, Suresh (Chandrasekar et al., 2013) interestingly, have compared their experimental results with the referent PV panel (non-cooled). According to their results, an electrical efficiency improvement as well as temperature reduction, varied based on the coolant used, but with water, it enhanced by 10.4% and reduced temperature by almost 20 °C (30%) respectively. The preparation procedure for nanofluids was not well elaborated and which is critical feature for long term stability of the produced nanofluids. The economic and environmental evaluation was also not obtained. According to another experiment by Owhaib, Qanadah (Owhaib et al., 2021), electrical efficiency improved up to 10% by a cold water filled clay pot. They used a water jacket shaped at the backside of the PV panel and the cold water from the clay pot naturally circulated through them, due to the thermosiphon effect, and by evaporation decreased the surface temperature of the PV panel. The proposed approach was tested only in the laboratory conditions, therefore real-field investigation should be provided. The conceptual design is also not fully suitable for real-field applications in the case of large-scale PV systems.

Additionally, Shahverdian, Sohani (Shahverdian et al., 2021a) have compared water flow cooling efficiency outcomes on two types of mono and polycrystalline PV cells with no cooling system (referent) and according to their results, although both panel showed significant improvement in electrical generation, monocrystalline cells outperformed. This result was somehow expected since monocrystalline PVs are in general more efficient. The optimum amount of water flow was also measured at around 0.01nullkg/s. They have highlighted the fact that water cooling is more suitable for the hot areas as it performs better in warmer environments. On the other side, the evaporation effect was not analyzed, and which is important aspect as water is limited resource on some locations. Table 3 illustrates in detail outputs of investigated research studies that have considered water-cooling strategy.

3.1.1.3. Nanofluid cooling. Nanofluids, represents the mixture of nanoparticles with water or other base fluids to enhance the thermal conductivity of the base fluid, are one of the novel cooling materials with an strengthened thermal properties (Yazdanifard et al., 2017; Jakhar et al., 2021). For instance, the electrical efficiency of the PV panel employing Al2O3-water Nanofluid is 1% more comparing to the pure water (Khanjari et al., 2017). So-called hybrid Nanofluids, referring to the mixture of more than one nanoparticle, seem to be a pioneer working fluid in the future (Kumar et al., 2020; Lari and Sahin, 2017). These particles could be added in various forms, weights, or percentages to improve the heat transfer (Rostami et al., 2018; Sardarabadi and Passandideh-Fard, 2016) That is to say, they are extensively suitable and expandable in the solar industry with wide application for PV panels as an optical filter or proficient coolant both in active and passive cooling systems (Chehreh, 2021; Said et al., 2018). However, utilizing Nanofluids makes the cooling systems more complex and expensive, despite raising the overall efficiency (Lari and Sahin, 2017). According to numerous studies (Chandrasekar et al., 2013; Su et al., 2021) and their results, in some cases, the electrical efficiency may decrease by adding nanoparticles comparing with pure water, whereas the total efficiency (thermal and electrical) is reinforced. In general, the type of nanoparticles and preparation process strongly affects thermal properties of

Table 3
Highlights of photovoltaic system passive water cooling studies.

Research	Year	PV cells	System	Technique	Temperature decrease	Electrical efficiency gain	Climate	Location
Shahverdian, Sohani (Shahverdian et al., 2021a)	2021	Polycrystalline Monocrystalline	PV	Water flow cooling system	NG NG	Efficiency: 3.84% Efficiency: 4.20%	Hot and dry	Tehran, Iran
Owhaib, Qanadah (Owhaib et al., 2021)	2021	NG	PV	Evaporative and thermosiphon effect using clay pot	The average does not exceed 42 °C	10%	NG	Jordan
Belyamin, Fulazzaky (Belyamin et al., 2021)	2021	Polycrystalline	PV	water flow cooling (WFC)	5.9–13.2°C-average = 10.1 °C	Depending on water flow rate, max electrical power from 9.5% to 11.9%	Tropical	Depok, Indonesia
Chandrasekar, Suresh (Chandrasekar et al., 2013)	2013	monocrystalline	PV	Evaporative effect using cotton wick structures	20 °C (30%)	Max electrical efficiency raised from 9% to 10.4%	tropical	Tiruchirappalli, India

nanofluids or nanocomposites (specific heat capacity, thermal conductivity, latent heat, etc.). Namely, in most of the cases thermal properties are usually being improved by addition of nanoparticles, while in some cases thermal properties could be even degraded. The addition of the nanomaterial would improve thermal conductivity, while specific capacity could be improved or degraded and latent heat is mostly being slightly reduced (Nižetić et al., 2020). The use of nanomaterial is also critical from toxicity aspect and safety in general.

Nonetheless, studies have indicated the fact that the heat transfer in this technique depends on various factors, namely, nanoparticle size, nanoparticle material, the geometry at the rear of the PV panel, ambient conditions, base fluid, etc. Moreover, the preparation method of the Nanofluid plays a key aspect that affects long-term stability of properties (Kumar et al., 2020; Nižetić et al., 2020). Among different types of nanoparticles, such as magnetic-based, carbon-based and metal-based, the former category seems more common and sliver (Ag) Nanofluids have recorded successful results (Lari and Sahin, 2017; Saidur et al., 2011). These outcomes are presented in Table 4 precisely.

3.1.1.4. Heat pipes. A heat pipe is a vapor-liquid phase change device which evaporates and expands (or rises, depends on the variant) on the one side of the PV panel, taking up the extra heat and releasing it into the surrounding when it condensates again on the other side and in a complete cycle, it circulates back as a liquid via capillary tubes (Grubišić-Čabo et al., 2016). System efficiency is clearly determined by the heat transfer capacity of the pipe material (Wu et al., 2011). Although, it is a costly device with complicated design techniques, it has high heat capacity and efficiency while being compact (Shittu et al., 2019a). Several parameters are effective in the performance of heat pipe cooling technologies, including working fluid, filling ratio, inclination angle, etc. (Alizadeh et al., 2018). The correlation between these mentioned parameters should be more investigated.

Moreover, heat pipes' array and configuration are essential subject in this field of research and studies have investigated different forms and shapes of heat pipes, namely single turn pulsating, flat plates, looped, etc. For example, Aamir Al-Mabsali, Candido (Aamir Al-Mabsali et al., 2021) arranged the heat pipes in a double-sided condenser span-wise with 50 mm spacing in the center and 3° inclination angle in a hot and arid climate, resulting in 29% enhancement in power performance efficiency. The proposed design is quite complex and it would be beneficial to discuss potential limitations of examined setup in order to better understand potential applicability of the solution in other climate circumstances. Alizadeh, Nazari (Alizadeh et al., 2020) have compared flat plate closed-loop pulsating heat pipe (CLPHP) in active and passive mode in four different scenarios, resulting in 23% and 35% maximum improvement in efficiency, respectively, indicating better electrical efficiency gain by the active technique. They have also highlighted the fact that only the statistical efficiency results are not reliable, by employing economical evaluation of life payback, and evaluating the operational costs, suggesting more cost-effective scenarios. The study is only missing an environmental evaluation in order to understand overall impact of the proposed concept.

Through a three-dimensional numerical evaluation of a flat plate heat pipe photovoltaic-thermoelectric (PV-TE) system, Shittu, Li (Shittu et al., 2019a) highly recommended this cooling method for HCPV systems because of its surpassing performance. However, the use of thermoelectric elements can be costly, therefore an economic evaluation is mandatory for the examined concept. The authors (Shittu, Li (Shittu et al., 2020)) have also demonstrated the feasibility of hybrid PV-TE with heat pipe for hot water production and power generation. They also highlighted a trade-off between increasing power production or efficiency and collecting supplementary hot water as a by-product through another experimental study. More details of the outputs of previously mentioned studies are illustrated in Table 5.

3.1.1.5. Liquid immersion cooling. In this method, a reasonably high-efficiency improvement could be obtained by submerging the PV panel (or in some cases cells) in a particular fluid with especial properties, as the excess heat can be taken away from both sides of the PV panel instead of just one surface (Gharzi et al., 2020). Therefore, this technique is particularly effective and widely studied both for concentrated photovoltaics (CPV) and in the cases when convectional PV panel was submerged into the water (usual cooling fluid in that case). Besides heat transfer properties, other criteria such as refractive index, suitable optical transmittance and optical stability should be considered to opt the best liquid, which are verified in water, de-ionized (DI) water, isopropyl alcohol (IPA), glycerol, ethyl acetate, dimethyl silicone oil, and etc. (Han et al., 2016). Another critical factor is the durability of the system in liquid environments and long-term degradation in these circumstances. Finally, the critical aspect of this technique is related to the presence of optical losses, and which are the most emphasized in for the higher thickness of the specific fluid (coolant). The presence of the optical losses is being affecting overall performance of the PV panel; therefore, this aspect should be carefully taken into account when considering specific design.

According to Sun, Wang (Sun et al., 2014)and Han, Wang (Han et al., 2016), de-ionized water seems unsuitable for multi-junction CPV solar cells due to the substantial absorption in the longer wavelength range, and dimethyl silicone oil was identified as the most stable liquid with no degradation in 180 days (about 6 months). The research mentioned could be further upgraded by thermal cycling tests in order to get more precise data related to the long-term stability of the cooling liquids.

The depth in which the PV panel are installed is another factor affecting the system's overall efficiency and main reason are optical losses as already noted. For instance, Sivakumar, Navakrishnan (Sivakumar et al., 2021) have experimentally investigated the efficiency of the solar systems in four various water depths and the best efficiency improvement was observed at 22 mm underwater and surpassing the 20 mm depth results in an efficiency decrease. According to the results, the electrical efficiency has an adverse relation with the depth of immersion as well. The water quality should be also investigated and its impact on the cooling performance. Moreover, for these kinds of concepts an environmental evaluation is crucial one.

Natural or built landscape marks namely canals, dams, lakes, rivers, seas, oceans, etc. could be a good place for floating or submerging large-scale PV panel underwater (Dwivedi et al., 2020; Mehrotra et al., 2014). However, this method may not be sustainable, as it affects visual parade of natural water landscapes and may threaten the life of the aquatics (Gharzi et al., 2020). Moreover, it is also not clear what is the long-term environmental impact of the floating PV plants in seas, rivers, or lakes, i.e., potential toxicity should be investigated. Nonetheless, this technique is less studied and documented (Kjeldstad et al., 2021). However,

Table 4
Highlights of photovoltaic system passive nano-fluid cooling studies.

Research	Year	System	Technique	Nanoparticle	Temperature decrease	Electrical efficiency	Climate	Location
Su, Sui (Su et al., 2021)	2022	CPVT	Simulations with a sub-continuous NDLBM	Al2O3-water	1.8–2.4 °C	0.7% drop	NG	NG
Chandrasekar, Suresh (Chandrasekar et al., 2013)	2013	PV	Evaporative effect using cotton wick structures	Al2O3-water CuO-water	17% 11%	9.7% 9.5%	tropical	Tiruchirappalli, India

Table 5
Highlights of photovoltaic system heat pipe cooling studies.

Research	Year	System	PV cells	Working fluid	Technique	Temperature decrease	Electrical efficiencygain	Climate/ weather	Location
Al Hasnawi, Judran (Al Hasnawi et al., 2021)	2021	PV	Polycrystalline	NG	heat pipe	7.5 °C	3.3%	Hot and dry	Baghdad, Iraq
Aamir Al-Mabsali, Candido (Aamir Al-Mabsali et al., 2021)	2021	PV	Polycrystalline	water	heat pipe heat exchanger (HPHE)	NG	29%	hot and arid	Muscat, Oman
Shittu, Li (Shittu et al., 2020)	2020	PV-TE	Crystalline silicon	water	flat plate microchannel heat pipe	6 °C	Overall efficiency: 12.19%	_	United Kingdom
Alizadeh, Nazari (Alizadeh et al., 2020)	2020	PV	NG	Distilled water	flat plate closed-loop pulsating heat pipe (CLPHP)	4–14.6 °C	3.5%, 7.5%, 23% depending on the irradiance	hot and arid	Kerman, Iran
Alizadeh, Ghasempour (Alizadeh et al., 2018)	2018	PV	Monocrystalline	NG	single turn pulsating heat pipes (PHPs)	16.1 °C	18%	_	_
Habeeb, Mutasher (Habeeb et al., 2017)	2017	PV	Monocrystalline	Distilled water 55%	thermosyphon heat pipe	15–35%	11–14%	Hot and dry	Baghdad, Iraq

Kjeldstad, Lindholm (Kjeldstad et al., 2021) assessed the cooling effect of a floating membrane with horizontally mounted PV panels in a river in Norway. Based on their evaluations, the efficiency improvement depends on the ambient and water temperature and this method outperforms the air-cooling technique. Again, as it was noted in the majority of the reviewed studies, the economic and environmental evaluation is missing. Table 6 is an illustration of the significant research outputs of examined studies.

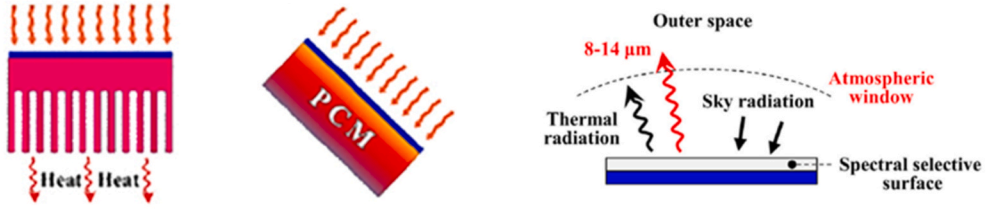
3.1.2. Conductive cooling

3.1.2.1. Heat sinks and wick structures. Implementation of additional conductive components in the forms of metals for cooling purposes could be in the general form of heat sinks, i.e., fins, wick structures, or a combination of these. Use of specific metals, due to their high thermal conductivity can intensify the heat transfer from the e PV panel to the surroundings and finally to maintain temperature of the panel at suitable level (Teo et al., 2012; Wu and Xiong, 2014; Firoozzadeh et al., 2019; Parkunam et al., 2020). Both cooling techniques are applied to the backside surface of the panel and owing to their simplicity and low cost are reasonable approaches for cooling purposes (Dwivedi et al., 2020; Wu and Xiong, 2014; Firoozzadeh et al., 2019). Heat sink refers to metal elements of specific geometry made of copper, iron and aluminum fins, which sometimes could be joined with wick structures c\ for more efficient air circulation (as shown in Fig. 4.) to ensure better cooling performance (Sato and Yamada, 2019; Salih et al., 2015). Research has been done broadly to investigate various metal sinks or to reach the optimum array of wick structures and fins' material. In study Parkunam, Pandiyan (Parkunam et al., 2020) authors concluded that the electrical efficiency of the copper fins is higher than the aluminum by 4% and 6%, respectively and which can be expected due to higher thermal conductivity of the copper. In the same study it is not clear why specific fin geometry and configuration was selected, i.e., on which basis. However, when considering the use of suitable metals for the fin-based cooling an economic aspect should be also considered. For instance, the market price of the copper is higher than aluminum and which can affect the overall economy. Moreover, the contact between the backside surface of the solar panel (Tedlar® layer) and the fins should be carefully designed to secure suitable heat transfer and to prevent heat losses.

Popovici, Hudişteanu (Popovici et al., 2016), through a numerical study, have investigated composition of the heat sink cooling capacity by adjusting the angle between the plate and the ribs, in which the maximum electrical efficiency was captured at 90°. They have proposed a low cost and energy-efficient method for cooling purposes; however, experimental validation is needed since investigation was only done by numerical way. On the other side, the effectiveness of the mentioned method strongly depends on the surrounding circumstances and general planned setup of the specific PV system. In another research, Ahmad, Fazlizan (Ahmad et al., 2021) numerically compared the performance of a multi-level fin heat sink (MLFHS) with that of a rectangular plate-fin heat sink by measuring fin spacing, fin height, and fin thickness. Based on their results, first the average temperature declines and then rises with fin spacing, fin height, and fin thickness. Amr, Hassan (Amr et al., 2019), in an experimental and theoretical study, have also drawn the same conclusion, by increasing the fins height and number, the electrical efficiency enhances, and it is influenced by the fins' thickness. The study didn't involve an economic evaluation, which would be an important benefit. Concerning fins' shape and configuration, Amber, Akram (Amber et al., 2021) have experimentally compared rectangular fins and circular fins' cooling performance, in which the rectangular fins were dominant both in electrical and thermal efficiency. It is not clear why and on which basis specific fin configuration was selected. Experimentally investigating cooling performance of fins, Čabo, Nižetić (Cabo et al., 2020) evaluated a PV system equipped with perforated aluminum fins attached on the back surface of the panels for 36 days, which was the first thorough analysis and study in a considerable amount of time in

Table 6
Highlights of photovoltaic system liquid immersion cooling studies.

Research	Year	System	PV cells	Method (liquid)	Temperature decrease	Electrical efficiency gain	Climate/ wheather	Location
Sivakumar, Navakrishnan (Sivakumar et al., 2021)	2021	PV	polycrystalline	Water immersion	At 35–37 °C	At 13.95–15.54% depending on the immersion depth	NG	NG
Kjeldstad, Lindholm (Kjeldstad et al., 2021)	2021	PV	NG	Water floating	NG	+5–6%	cold	Skaftå, Norway
Mehrotra, Rawat (Mehrotra et al., 2014)	2014	PV	polycrystalline	Water immersion	At 31–39 °C	17.8%	Hot	Bhopal, India
Sun, Wang (Sun et al., 2014)	2011	CPV	monocrystalline	de-ionized water	5–16 °C	At 13.5%	sunny and clear	Las Vegas, USA



(a) Heat sinks (b) PCM cooling (c) Radiative cooling

Fig. 4. Schematic diagram of passive conductive and radiative cooling techniques (Gharzi et al., 2020; Sato and Yamada, 2019).

realistic condition. They reached almost 5% increase in power yield tied with the Global Tilted Irradiance (GTI), which means that gained cooling results could be directly weight against various locations globally. They have also included a life cycle assessment (LCA) to evaluate the PV system coupled with fins from environmental aspect. Their analysis suggest that the investigated technology will contribute to CO_2 emissions reduction after 7 years, and the rest of the operating lifespan will provide net CO_2 emissions savings. The conducted research was important since it was obtained in an integral way, i.e., proposed concept was evaluated from techno, economic and environmental aspect.

Rajput and Yang (2018) drew an experimental comparison between cylindrical pin fin heat sinks and single-channel PVT collectors. Based on their results, the single-channel PVT collector, which is considered the best cooling method, outperformed the heat sink. Although, their research lacks result of electrical efficiency, they have strongly suggested further investigation under stagnant wind conditions. The proposed research was obtained on the 20 WnullPV panel, and it would be beneficial to repeat investigation on the large-scale PV panel in order to determine performance response. Kim, Bae (Kim et al., 2020) have suggested metal meshes based on their research, despite being less efficient than fins, they are more likely to be adopted at a large scale due to their more straightforward production and lower cost. Precise results of the examined existing research studies are shown in Table 7.

3.1.2.2. PV-PCM cooling. PCMs or Phase Change Materials are specific substances with various melting points and suitable capacity for thermal energy retaining, with high latent heat of fusion by unit volume with ideally small thermal expansion coefficient. All previously mentioned, allows them to maintain considerable amount of thermal energy for temperature stabling (Kalaiselvan et al., 2018; Dwivedi et al., 2020). This in turn, causes significant release or absorption of energy throughout the phase changing process, melting or freezing (Dwivedi et al., 2020). Although, this routine cannot be considered as cooling in nature, it is capable of restraining temperature rise and its stabilization, making them a suitable method for passive cooling in PV systems (Grubišić-Čabo et al., 2016; Kalaiselvan et al., 2018; Dwivedi et al., 2020). By placing specific PCM material under PV panel (as shown in Fig. 4.), the extra heat from the cells can be absorbed throughout the day by melting down and releasing that to the surroundings throughout the night by consolidation (Kalaiselvan et al., 2018; Chehreh, 2021). However, the key issue regarding the use of PCM materials is aspects concerning the efficient thermal management of the PCM layer. Namely, if the heat transfer from the PCM layer to the surroundings is not an effective one, the PV panel would remain operating on the relatively high temperature. Due to previously mentioned issue, the design of the PV-PCM cooling systems should be carefully obtained and considering specific climate conditions and with necessary optimization of the PCM layer. Finally, an economic aspect of these materials is also important one since unit price of PCM materials is usually high (5 € to 15 €/kg) and strongly affects economy of the PV-PCM cooling design.

Variety of PCMs are accessible in forms of inorganic salt hydrates, organic oil and eutectics. The key factors for adopting proper PCM for passive cooling are the latent heat, the melting temperature, and the PV

Table 7
Highlights of photovoltaic system heat sinks and wick structures (fins) cooling studies.

Research	Year	System	PV cells	Material	Technique	Temperature decrease	Electrical efficiency	Climate/ wheather	Location
Ahmad, Fazlizan (Ahmad et al., 2021)	2021	PV	NG	aluminum	truncated multi-level fin heat sink (MLFHS)	6.13%	+2.87%	Tropical	Malaysia
Amber, Akram (Amber et al., 2021)	2020	PV	Monocrystalline	Rectangular Circular	Passive cooling with fins	10.6% Negligable	14.5%. Gain 13.2% Gain	_	Pakistan
Čabo, Nižetić (Čabo et al., 2020)	2020	PV	polycrystalline	aluminium	perforated fins on the back surface	varied	5% gain	Mediterranean	Croatia
Kim, Bae (Kim et al., 2020)	2019	PV	NG	Iron Aluminum	Cooling with iron and aluminum meshes	4.35 °C 6.56 °C	15.51% 16.85%	_	NG
Amr, Hassan (Amr et al., 2019)	2019	PV	crystalline	NG	Experimental and theoretical study of metalic fins	4–5 °C	13.4–13.86%	Hot and dry	El-Minia Egypt
Parkunam, Pandiyan (Parkunam et al., 2020)	2019	PV	Monocrystalline	copper aluminum	heat sinks and wick structure with copper and aluminum fins	At 40.06 °C At 44.41 °C	12.52% 12.29%	_	Trichy, India.
Rajput and Yang (Rajput and Yang, 2018)	2018	PVT	polycrystalline	aluminum	cylindrical pin fin heat sinks with 1.22 fin*cm^{-2} density	Form 88.1 °C–58.4 °C	NG	NG	Ontario, Canada
Popovici, Hudişteanu (Popovici et al., 2016)	2016	PV	NG	copper or aluminum	ribbed wall heat sink with air cooling	>10 °C	7.55%	clear day of summer	_

panel material (Dwivedi et al., 2020). For instance, salt-based PCMs may cause corrosion in reaction with metal, whereas paraffin may react with plastics (Dwivedi et al., 2020). Finally, the specific desired temperature level of the PV panel, in certain climate circumstances, would also have an impact on selection of the PCM material. Although, these materials are not conductive in the nature, their appliance as a heat sink is in some ways similar to the previous category. Due to this method's inherent attributions and intense dependence on the ambient temperature, environmental factors play a critical role in this field of research and various studies have assessed PCM cooling performance in different weather conditions or geographical locations. Table 8 indicates the results of the following reviewed studies.

Sharaf, Huzayyin (Sharaf et al., 2021) have conducted a research in winter in duration of three months to compare aluminum metal foam (AMF)-PCM PV system with a referent PV panel (non-cooled one). According to their results, the cooling performance improves by a decline in ambient temperature, and they suggested investigations in summer for further research. The experimental work was obtained on the small scale PV panel (20 W) while performance response on the large scale PV panel could be rather different (the size of the PV panel is important one in the case of PV-PCM systems due to thermal inertia of the PCM layer). Hasan, Sarwar (Hasan et al., 2017a) highlighted the fact that most of the studies on cooling technologies have been conducted in a short-term investigation. Therefore, they have adopted a long-term approach in their research throughout a whole year. Based on their results, the PCM achieved less cooling effect in peak cool and peak hot months of the year because of its deficient melting and solidification, respectively. An improvement of the cooling performance in the previous case could be obtained by proper optimization of the PCM layer. The authors didn't provide an economic and environmental evaluation.

Moreover, the thickness of the PCM is another factor affecting PV panel performance response. Hassaboua, Abotaleba (Hassaboua et al., 2019) have concluded through experimental research for nine months that 20–30 mm is the optimum thickness for PCM sink in the outdoor condition of the research location in Qatar which has an extremely hot and humid climate which exacerbates the overheating of the PV system. Although, authors have compared performance of several cooling approaches with referent (no cooled) PV panel, the economic evaluation is missing, so it is hard to determine which cooling concept is the most suitable. In another research, which is actually an integration of PCM and pin fin cooling, Agarwal and Saxena (2021) studied the performance of multi-cylindrical copper tubes filled with PCM in three different configurations, 91 (B), 181 (C), and 271 (D) number of tubes to be compared with a reference (A) PV system with non-cooled, and the highest tube numbers obtained the best result. It is not clear how specific position and configuration of the tubes was selected, since proposed configuration could be sensitive in the case of low wind conditions, i.e., where weaker overall heat transfer could occur.

3.1.3. Radiative cooling

Former technologies and the followings in this review, have not yet been used widely on a large scale due to their complexity and challenges throughout installation and maintenance (Li et al., 2021a). Therefore, more straightforward modifications with no extra additional large components, such as radiative cooling, have drawn growing attention both in material science and engineering fields (Li et al., 2021a; Zhao et al., 2020). By applying a high emissive thin-film layer with a wavelength of 8–13 μm on the surface of the solar panel, an integrated radiative cooling is obtained (Dwivedi et al., 2020; Li et al., 2021a; Chehreh, 2021). Fig. 4 illustrates a schematic diagram of a radiative cooling configuration. The change in absorption and thermal radiation by the spectral emissivity of the solar system is mainly based on Nanotechnologies (Han et al., 2019). However, in order to prevent adverse effects on the efficiency of the PV panel, the transmissivity of the solar spectrum band should remain high, i.e., around 0.28–2.5 μm (Li et al., 2021a). Nowadays, the emergence of a novel field in optical science called spectral beam splitter made an advance in PV cells' coating technology. It is capable of filtering the solar irradiance infrared spectrum, which is the reason for temperature rise in PV systems (Sibin et al., 2017; Ju et al., 2017; Yin et al., 2018). Wang, Kortge (Wang et al., 2020), by coupling radiative cooling in the CPV system under a variety of wind speeds and solar heat loads, achieved significant results and increased the system's lifespan. However, the investigation was obtained on rather small experimental setup and with also lack of economic evaluation.

As this method is still in its early stage, controversial debates exist on its practical application and efficiency. Knowing the fact that the commercial solar cells in the market are already capsulated with glass containing 70% silica with an acceptable thermal emissivity rate cannot justify the use of this method (Li et al., 2021a; Zhao et al., 2018; Ahmed et al., 2021). For instance, Li, Ahmed (Li et al., 2021a) (Ahmed et al., 2021), through a couple of integrated experimental and numerical research, have compared the cooling performance of a typical glass capsulated solar panel with a panel coated by polydimethylsiloxane. They did not record a substantial and considerable enhancement in the performance of the solar systems. The economy and long-term stability of the proposed cooling approach is not clear. Similarly, Zhao, Hu (Zhao et al., 2020), in their review study on radiative cooling, have highlighted the fact that most studies had a bias towards the novelty of this approach with some overestimations on its effects, compared to encapsulated PV panel with almost the same performance. Zaite, Belouaggadia (Zaite et al., 2021) have also concluded from their study that night radiative cooling is less efficient for the glazed PVT collector, due to the adverse effect of glass on the infrared radiation emitted by the PV panel. More details of the presented studies can be found in Table 9. Despite the advantages of the radiative cooling technique, further investigation is needed to optimize its performance in comparison to commercial PV panel existing in the market, and in any way, there is room for further research and discussions (Sato and Yamada, 2019).

Table 8
Summarized features of PV-PCM cooling approaches.

Research	Year	System	PV cells	Material	Technique	Temperature decrease	Electrical efficiency gain	Climate/ weather	Location
Agarwal and Saxena (Agarwal and Saxena, 2021)	2021	PV	Polycrystalline	Copper tubes, PCM: NG	multi-cylindrical copper tubes filled with PCM	20.76%	+6.67%	NG	Moradabad India
Sharaf, Huzayyin (Sharaf et al., 2021)	2021	PV	NG	Paraffin based	aluminum metal foam (AMF) with PCM	Dec:4.3% Jan:8% Feb:15%	Dec:+1.85% Jan:+3.38% Feb:+4.14%	Winter	Benha, Egypt
Hassaboua, Abotaleba (Hassaboua et al., 2019)	2019	PV	Mono, poly crystalline	NG	Passive PCM cooling	10 °C	At 9–11%	Extremely hot and humid	Qatar
Hasan, Sarwar (Hasan et al., 2017a)	2017	PV	Polycrystalline	Paraffin based	Passive PCM cooling	10.5 °C	+5.9%	extremely hot	UAE

Table 9
Highlights of photovoltaic system radiative cooling studies.

Research	Year	System	PV cells	Technique	Temperature decrease	Electrical efficiencygain	Climate/ wheather	Location
Li, Ahmed (Li et al., 2021a)	2021	PV	Monocrystalline	Coating by polydimethylsiloxane (PDMS)	<2 °C	+0.76%		Shanghai, China
Zaite, Belouaggadia (Zaite et al., 2021)	2021	PVT	Polycrystalline	night radiative cooling technology	3–5 °C	+5.5%–6.15%	arid	Ouarzazate, Morocco
Wang, Kortge (Wang et al., 2020)	2020	CPV	GaSb cell	coating with LDPE film	5–36 °C	8–27%	Various conditions	NG

3.1.3.1. Photonic devices. The use of periodic dielectric crystals in photonic cooling research and in the provision of emittance spectra and photovoltaic cooling information has shown high potential (Maka et al., 2003). Photonic crystals are periodic dielectric structures used to design the energy band structures for photons and allow or restrict propagation of electromagnetic waves of certain frequencies (Maka et al., 2003). Despite being a robust strategy to enhance the efficiency of PV systems (Zhang et al., 2021a) there exists substantial challenges and issues in the field. Namely the prices, manufacturing, design, etc. needs to be addressed before this technology can be deployed in real-world applications (Zhang et al., 2021a).

To the best of our knowledge, research in this field is mainly limited to numerical or modeling studies. For instance, Zhu, Raman (Zhu et al., 2014) utilized the rigorous coupled wave analysis (RCWA) technique to simulate electromagnetic (EM) structure of micro photonic cooling strategy. Or, Perrakis, Tasolamprou (Perrakis et al., 2020) employed an electrical-thermal co-model to examine photonic strategies for controlling heat generation in PV systems through ultraviolet and sub-bandgap reflection. According to their findings, the most efficient method for reflecting sub-bandgap radiation is utilizing the PV panel as a parasitic absorber with a high parasitic absorption. Additionally, their study demonstrated that the reflection of UV radiation may decrease the operating temperature of the PV panel and enhance their efficiency. Similarly, Gao, Xia (Gao et al., 2020) simulated spectrally selective reflector using photonic structures combined with spectrum-selective mirrors in different meteorological conditions. Their passive cooling technique recorded significant results in both low and high wind speed with superior performance. In all previously mentioned studies, the key issue is that investigation was only provided by numerical way, i.e., there is need for experimental evaluation in order to understand realistic potential and to determine specific technological limitations. Moreover, the discussion of economic aspects is also beneficial in this field.

From an experimental perspective, Zhao, Lu (Zhao et al., 2022) proposed silica micro-grating photonic cooler which has slight anti-reflection effect for sunlight as well as being transparent. It needs to be noted that temperature reduction of PV panel in the mentioned study was modest, i.e., around 3.6 °C and which makes in question the economic viability of the proposed cooling method. Accurate results of the reviewed studies could be found in Table 10. It worth mentioning, the proposed cooling method by photonic devices is still in the phase of intense development and more research efforts should be provided in order to understand overall benefits of the considered cooling approach.

3.2. Active cooling

Active cooling techniques consume additional operational energy (usually electricity) to maintain fluid circulation by pumps or fans, ultimately decreasing the temperature of the PV panel in certain extent. Despite being more efficient than the passive methods, specifically in hot climates and arid regions like deserts (Gharzi et al., 2020; Schiro et al., 2017), economic aspect requires consecration of power consumption to production (Dwivedi et al., 2020). That is to say, a self-sufficient cooling system is needed to support its power consumption, which is not a severe problem with concentrated PV cells due to the ability to use less cooling fluid and the fluid-to-cell mass ratio (Grubišić-Čabo et al., 2016).

3.2.1. Forced flow

Using water or air as a coolant, a pump or fan ensures forced and more effective circulation over the PV panel or PV system. These attached types of equipment could be sprays, fans, pumps, turbines, etc., which in this part are categorized based on their coolant, namely water, air, or even Nanofluids. All these categories are illustrated schematically in Fig. 5 in the following. The key aspect here is selection of the fluid, due to different thermal properties of various coolants and to also consider specific configuration.

3.2.1.1. Air-based. Through the active air-cooling method the forced airflow is employed to enhance the heat transfer (Gharzi et al., 2020; Mazón-Hernández et al., 2013). Research outcomes of the reviewed air-based cooling studies for PV panels are presented in Table 11. Several studies have been conducted to determine different installations, best configuration, and arrangement for optimized and uniform cooling performance. For instance, Shrivastava, Jose (Shrivastava et al., 2022) conducted an experimental research to identify optimal configuration fins, ducts, and baffles to enhance cooling performance. In order to do so, they tested different arrays of ducts and fins which the longitudinal fins with inclined baffles recorded the best performance. Their research highlighted that heat transfer coefficient dramatically rises with an increase in Reynolds number. Proposed installation also had exergy enhancement on the thermal system. The economic and environmental evaluation was not discussed in the mentioned work, which is beneficial. Similarly, through a numerical simulation, Alsayah, Aboaltabooq

Table 10
Highlights of photovoltaic system photonic devices cooling studies.

Research	Year	System	PV cells	Technique	Temperature decrease	Electrical efficiencygain	Climate/wheather	Location
Zhao, Lu (Zhao et al., 2022)	2022	PV	NG	Silica micro-grating photonic cooler	3.6 °C	NG	NG	NG
Perrakis, Tasolamprou (Perrakis et al., 2020)	2020	PV	monocrystalline	Phtonic strategies: ultraviolet and sub-bandgap reflection	up to 10 °C	up to 5.8%	_	_
Gao, Xia (Gao et al., 2020)	2020	PV	crystalline silicon	Photonic structures based spectrum-selective mirror	23.2–68.1 °C	0.5–6.7%	Different meteorological conditions	_
Zhu, Raman (Zhu et al., 2014)	2014	PV	crystalline silicon	Radiative cooling using 2D microphotonic thin film	18.3 °C	7.9%	_	_

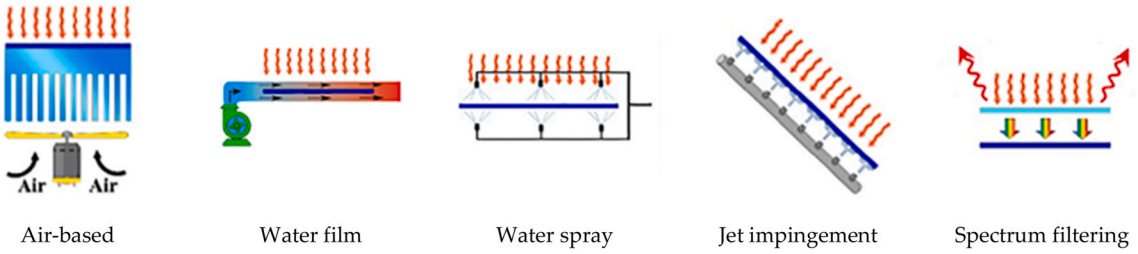


Fig. 5. Schematic diagram of active forced flow cooling techniques (Gharzi et al., 2020).

Table 11
Highlights of photovoltaic system air-based forced flow cooling studies.

Research	year	System	PV cells	Method	Temperature decrease	Electrical efficiency gain	Climate	Location
Shrivastava, Jose (Shrivastava et al., 2022)	2022	PVT	NG	Forced air cooling with ducts and fins	At 65-56 °C	5–7%	Hot	Chennai, India
Li, King (Li et al., 2021b)	2021	PV	Monocrystalline	compressed air-based regulation	At 40–50 °C	30.7–36.1%	arid	India
Maghrabie, Mohamed (Maghrabie et al., 2017)	2017	PV	NG	Forced air cooling using air blower and ducts	At 42.1 °C	+3.7%	NG	Sohag, Egypt
Amelia, Irwan (Amelia et al., 2016)	2016	PV	NG	DC fan	22.22%	37.17%,-44.34%	Hot	Malaysia

(Alsayah et al., 2019b) tried to find out the optimum array of air guides in forced airflow cooling and proposed it in aluminum channels fixed to the bottom of the panel. The dimension, angle and quantity are suggested to be opted regarding the dimensions of the PV cell referring to the same study. Study requires experimental verification.

Research has been done to investigate other modifications or potentials of forced air cooling. Li, King (Li et al., 2021b) studied the cleaning potentials of the compressed air-based regulation for dust particles. Based on their results, to balance energy consumption for air compressing and energy gained from PV performance the blowing time and specific particle size for removal should be considered. On the other hand, through their experimental research, Amelia, Irwan (Amelia et al., 2016) investigated the number of fans to be used for PV system cooling. In that sense the optimization approach regarding the suitable number of fans should be provided, together with economic evaluation.

3.2.1.2. Water-based. Owing to the more uniform distribution on the surface, potential of light absorption enhancement, high thermal capacity, cleanness and extended system lifespan, active water flow cooling (WRF) is among the most popular and suitable techniques. However, the energy consumption, performance optimization and approach have not been fully addressed in existing research works (Shahverdian et al., 2021b). However, they are some positive examples of realized active cooling system and that have both performance as well as economic aspect (Castanheira et al., 2018). Water-based forced flow techniques include technologies involving evaporative cooling, thin-film layer, or water spray for instance. Important feature that should be considered when water is considered as coolant is evaporation effect and that affects water consumption as already mentioned. Moreover, there are issues with the water shortage in some regions of the world which should also be taken into account.

Different studies have investigated active water-cooling methods in different climates concerning various factors in this technique and optimization aspect. According to Bahaidarah, Subhan (Bahaidarah et al., 2013) and Abdullah, Ping (Abdullah et al., 2021), a rise in the mass flow rate of water leads to reduce in temperature rise. The cooling designs in mentioned were rather simple. Nevertheless, by implementing dynamic multi-objective optimization (DMOO) on a water-based cooling system, Shahverdian, Sohani (Shahverdian et al., 2021b) enhanced the performance of the PV system significantly by maximizing the electrical efficiency up to 64.73% and minimizing water consumption by 41.98%. Table 12 illustrates the results and methods of these studies.

In the water spray technique, the water is sprayed on either the front or back surface of the panel through the pump the connected pipes (Dwivedi et al., 2020). However, studies have shown the front side spraying is to be more efficacious (Moharram et al., 2013) however there is not major difference in the performance when considering front or backside cooling (Nižetić et al., 2016). Although, according to Nižetić, Čoko (Nižetić et al., 2016) simultaneous spraying on the both sides of the panel has the best cooling results, It is actually mentioned that there is an approximate linear dependency between the water spray flow influence and electrical efficiency. Table 13 presents the highlights of water spray cooling studies. According to Yesildal, Ozakin (Yesildal et al., 2022), spraying time, spray flow rate and solar irradiation are the most effective factors on the optimization of the PV panel in water spray cooling. They applied air-assisted water spray on the PV panel surface, and they reached the optimum values for higher electrical efficiency at higher spray flow rate, lower nozzle air flow rate, and mean spraying time values. Similarly, to optimization ends, Hadipour, Zargarabadi (Hadipour et al., 2021) proposed a pulsed-spray water due to less water consumption in comparison to steady water sprays, and they decreased the water flow rate by one-ninth through this approach while increasing the electrical efficiency by 27.7%, with also reducing the costs about 46.5%–76.3% with different currents. Through another experimental research, Yang, Liang (Yang et al., 2019) integrated spray cooling with a shallow geothermal energy heat exchanger to enhance the efficiency of the PV system. Based on their outcomes, the electrical efficiency raised by 14.3% powered by 10 panel and the initial investments will compensate in 8.7 years. According to Nižetić, Čoko (Nižetić et al., 2016) and based on their experimental study, this technique also acts as a booster to the average electricity production because of its self-cleaning effect. It needs to be noted that when considering all herein mentioned water spray-based cooling studies, there is lack of more detail analysis of the evaporation effect in different climates. This aspect is very important to estimate water loss due to evaporation, and which can be critical factor in specific geographical locations with limited water supply.

Water film cooling, among all active cooling methods, is one of the best solutions to provide uniform temperature distribution while having

Table 12
Highlights of photovoltaic system water-based forced flow cooling studies.

Research	year	System	PV cells	Method	Temperature decrease	Electrical efficiency gain	Climate	Location
Shahverdian, Sohani (Shahverdian et al., 2021b)	2021	PV	Polycrystalline	dynamic multi-objective optimization (DMOO) of water flow cooling (WFC)	54.07%	64.73% annually	Hot and dry	Tehran, Iran
Abdullah, Ping (Abdullah et al., 2021)	2021	PV	Polycrystalline	Using cooper pipe and pump	16 °C	+3%	NG	_
Bahaidarah, Subhan (Bahaidarah et al., 2013)	2013	PV	Monocrystalline	Heat exchanger (cooling panel)	20%	+9%	Hot and dry	Dhahran, Saudi Arabia

Table 13
Highlights of photovoltaic system water spray cooling studies.

Research	year	System	PV cells	Method	Temperature decrease	Electrical efficiency gain	Climate	Location
Yesildal, Ozakin (Yesildal et al., 2022)	2022	PV	NG	air assisted water spray	NG	At 9.4%–11.435%	_	Laboratory
Hadipour, Zargarabadi (Hadipour et al., 2021)	2021	PV	Monocrystalline	a pulsed-spray water	At 24.8 °C - 26.5 °C	25.9%–27.7%	Hot and dry	Semnan, Iran
Yang, Liang (Yang et al., 2019)	2019	PV	NG	Water spray equipped with heat exchanger	at 27–29 °C	At 14.3%	NG	NG
Nižetić, Čoko (Nižetić et al., 2016)	2016	PV	Monocrystalline	Water spray	30 °C	5.9%	Mediterranean	Corotia

a surface cleaning effect (Mah et al., 2019). This technique could also be implemented on the front or back surface of the PV panel (Gharzi et al., 2020) by covering the surface with a thin layer of water flow. The additional advantage of the water flow over the front surface is cleaning effect and that could be beneficial when PV systems are operating in harsh environments. Studies investigated in this review could be found in Table 14. Based on numerical modelling, Hudişteanu, Cherecheş (Hudişteanu et al., 2021) investigated the impacts of water film thickness on cooling performance as well as electrical efficiency of the PV system. The optimal solving was recorded for 3 mm of thickness water film, operating temperature of 20 °C and velocity of 0.01nullm/s. This study only dealt with numerical investigation and there is necessity for experimental verification. Similarly, Mah, Lim (Mah et al., 2019) investigated on optimal cooling water flow in tropical climate, which was found at 6nullL/min rate. It was also concluded that through this technique non-uniform temperature distribution on the surface of the panel was reduced. They also proposed recycled local rainfalls to cover the water consumption and that is very important to secure responsible water utilization. The mentioned use of different water resources for PV panel cooling, such as rainfall or recycled water is important one from sustainability aspect.

Finally, the most useful, pragmatic, and high-potential method for controlling excessive heat, evaporative cooling, which has the same mechanism for cooling the PV system as the sweat cools down the body, works on water evaporation with the surrounding air phase equilibrium (Dwivedi et al., 2020). This method is cost-effective, easy to implement and effective in dry climates, and can perform on stable and competitive way (Haidar et al., 2018, 2021). Kadhim and Aljubury (2020), experimented the application mentioned technique on four different cases of the front side, both sides, using Arduino controller with two different water flow rates and compare it with the non-cooled system (referent). The optimum outcomes were recorded from the panel with Arduino controller in which the circulation pump operation depended on the PV panel operating temperatures. The study didn't involve an economic evaluation, which is kind of a drawback. More results have also been presented in Table 15.

3.2.1.3. Jet impingement. The jet impingement technique involves forming a very thin thermal boundary layer on the back side of the PV panel by throwing a fluid at high flow velocity to accelerate the heat transfer process and minimize thermal resistance against contact surfaces and closed turbulent flow (Gharzi et al., 2020; Chehreh, 2021). In order to maintain a steady and uniform cooling performance, an array of jets are normally used (Nadda et al., 2018). It was also found that the efficiency of this technique is increased when accompanied with micro channels (Gilmore et al., 2018).

The performance of the jet impingement system is affected by circumstances namely nozzle pitch, the type of the working fluid, nozzle-to-plate spacing, geometry and numbers, Brandt and Reynolds numbers (Chehreh, 2021; Nadda et al., 2018; Hasan et al., 2017b). According to the obtained review, multiple arc protrusion on a plate is a better arrangement for impinging jets (Nadda et al., 2018). Investigating on the impact of the number of nozzles, their sizes (diameter), and the dimensionless nozzle-to-PV spacing on the overall performance, Javidan and Moghadam (2021) conducted an indoor study and based on their results, the optimum performance was achieved with an increase in the liquid mass flow rate, the number of nozzles and minimum nozzle-to-PV spacing for best uniformity. The investigation in the mentioned study was obtained in laboratory conditions and on the small PV panel (10 W), therefore the data regarding performance response should be carefully taken into account. Moreover, the optimum performance was achieved with a rise in the liquid mass flow rate (Torbatinezhad et al., 2021), the

Table 14
Highlights of photovoltaic system water film cooling studies.

Research	year	System	PV cells	Method	Temperature decrease	Electrical efficiency gain	Climate	Location
Hudişteanu, Cherecheş (Hudişteanu et al., 2021)	2021	PV	Monocrystalline	water film heat exchanger	At 22.3 °C	30.6%	_	_
Mah, Lim (Mah et al., 2019)	2019	PV	Silicon crystalline	water-film cooling	28 °C	15%	tropical	Malaysia.

Table 15
Highlights of photovoltaic system evaporative cooling studies.

Research	year	System	PV cells	Method	Temperature decrease	Electrical efficiency gain	Climate	Location
Haidar, Orfi (Haidar et al., 2021)	2021	PV	NG	evaporative cooling	10 °C	5%	Hot and dry	Riyadh, Saudi Arabia
Kadhim and Aljubury (Kadhim and Aljubury, 2020)	2020	PV	NG	Evaporative cooling equipped with geothermal energy	12.6 °C	Max 16%	hot and dusty	Baghdad, Iraq
Haidar, Orfi (Haidar et al., 2018)	2018	PV	NG	evaporative cooling	>20 °C	14%	Hot and dry	Riyadh, Saudi Arabia

number of nozzles and minimum nozzle-to-PV spacing for best uniformity. Hasan, Sopian (Hasan et al., 2017b) studied the effect of implementing different Nano particles (SiC, TiO2 and SiO2) in water as a coolant and based on their results, SiC/water recorded the best cooling performance. The use of jet impingement technique in combination with nanofluids could be critical from safety point of the view, therefore this investigation requests a necessary environmental evaluation. Details of jet impingement cooling technique studies could be found in Table 16.

3.2.1.4. Spectrum filter cooling. In this method, as shown in Fig. 6., the PV panel is thermally shielded by liquids such as Nano-fluids, organic liquids or vapor which can act as optical filters to deliver only the useable spectrum for power production (Dwivedi et al., 2020). Liquid filters can circulate constantly with magnetic/electric fields, because of the decline in the band gap energy of the semiconductor material at the higher cell temperature, and temperature alters over specific solar wavelengths from ultraviolet to near-infrared (Dwivedi et al., 2020; Kabeel et al., 2021). This cooling method is broadly accompanied with the hybrid solar photovoltaic-thermal applications (Kabeel et al., 2021). Transparent fluids such as ethylene glycol, glycerin, silicone oil, coconut oil, etc., could be used as optical filters (Joshi and Dhoble, 2017, 2018), among which, based on current research findings coconut oil performs better than water and silicone oil (Dwivedi et al., 2020; Joshi and Dhoble, 2017). Table 17 illustrates the highlights of the spectrum filtering studies investigated in this chapter. Joshi and Dhoble (2017) have compared the performance of three distinct fluids as spectrum filters using tap water, coconut oil silicone oil. According to their results, coconut oil and silicone oil showed UV absorption and IR region. On the other side study didn't consider an economic aspect.

However, through an experimental study on spectral-splitting using Ag Nano fluid in a PVT system, Zhang, Shen (Zhang et al., 2021b) concluded a decrease in electricity yield but an increase in the harvest of heat. The experimental approach was conducted on the small PV panel, which is limitation and preparation approach for nanofluids was also not well elaborated.

3.2.2. Heat sinks

It consists of transferring heat from one type of medium to another, or from one liquid to another, using the specific design of heat exchanger (Dwivedi et al., 2020). Waste heat could also be harvested for secondary purposes in PV systems coupled with thermal ones (Gilmore et al., 2018). Different designs and configurations of heat sink active cooling techniques can be found in Fig. 6.

3.2.2.1. Heat exchangers. Heat exchangers could be in form of microchannel fixed to the back side of the PV panel. Depending on their application, microchannel heat sinks are made up of various geometries such as trapezoidal, triangular, rectangular, and circular channels of different sizes, between 10 and 3 μm, depending on their general application (Al-Asadi et al., 2022). Micro-channels are typically categorized as two-phase or single-phase regarding their use of homogenous or non-homogenous fluids (Al-Asadi et al., 2022). However, Radwan, Ookawara (Radwan et al., 2019) concluded from their studies that with two-phase flow boiling, higher cooling efficiency could be achieved at lower flow rates in comparison to single-phase liquid flow at the same flow rates. Moreover, according to Radwan, Ookawara (Radwan et al., 2018b), the average temperature and uniformity will improve by intensifying the flow rate. In that sense there is lack of sufficient number of optimization studies in this field and which would be beneficial for economic aspect.

The design and arrangement of variant micro channels configurations are critical topics in this field of research which are presented in Table 18. For instance, Radwan and Ahmed (2017) compared the performance of different micro channel configurations, namely a single layer parallel- and counter-flow microchannel, wide rectangular microchannel, and a double layer parallel- and counter-flow microchannel through a comprehensive, three-dimensional thermo-fluid model. According to the results, a single layer parallel-flow microchannel configuration outperformed in terms of the highest cell net power, electrical efficiency, and the minimum cell operating temperature and in contrast, a single layer counter-flow microchannel configuration recorded the least optimization influence. The study was only directed to the performance analysis of various microchannel cooling setups; however, the economic evaluation is missing. Multi-layered microchannel among many other techniques, are more efficient owing to better heat removal and pumping power (Al Siyabi et al., 2018). Hence, Al Siyabi, Khanna (Al Siyabi et al., 2018) examined its performance on a CPV system both experimentally and numerically and based

Table 16
Highlights of photovoltaic system jet impingement cooling studies.

Research	Year	System	PV cells	Working fluid	Technique	Temperature decrease	Electrical efficiency gain	Climate/ wheather	Location
Javidan and Moghadam (Javidan and Moghadam, 2021)	2021	PV	polycrystalline	water	multi-nozzle jet impingement cooling	30.27 °C	47.67%	indoor	_
Torbatinezhad, Rahimi (Torbatinezhad et al., 2021)	2021	CPV	NG	water	jet impingement/ mini-channel heat sink	Comparing points for uniformity	28.95%	NG	NG
Radwan, Hawi (Radwan et al., 2018a)	2018	CPV	Polycrystalline	water	jet impingement heat sink	20–30 °C	NG	NG	NG
Hasan, Sopian (Hasan et al., 2017b)	2017	PVT	Polycrystalline	Nanofluids (SiC/water)	nanofluids jet impingement	35–46 °C	At 12.75%	indoor	Malaysia

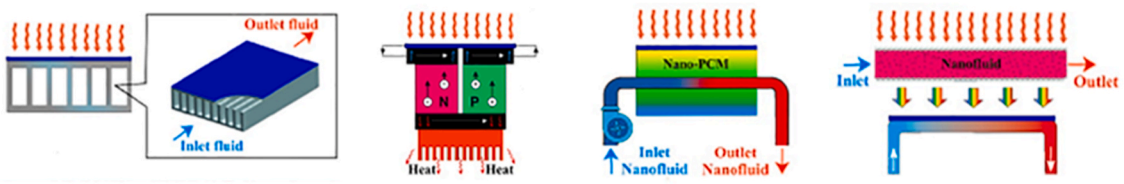


Fig. 6. Schematic diagram of heat sinks and spectrum filter cooling techniques (Gharzi et al., 2020).

Table 17
Highlights of photovoltaic system spectrum filtering cooling studies.

Research	Year	System	PV cells	Working fluid	Technique	Temperature decrease	Electrical efficiencygain	Climate/ wheather	Location
Zhang, Shen (Zhang et al., 2021b)	2021	PVT	crystalline silicon	Ag nanofluid	spectral-splitting	At 25 °C	2.12–2.78% decrease	indoor	China
Ling, Li (Ling et al., 2020)	2020	CPVT	CdTe	methanol	spectral filter	100 °C	up to 39%	NG	NG
Huaxu, Fuqiang (Huaxu et al., 2020)	2020	CPVT	Monocrystalline	SiO2/TiO2	optical splitting film-based spectral splitting	11 °C	9.4%	NG	Weihai, China
Liang, Han (Liang et al., 2019)	2019	PVT	Polycrystalline	SiO2/TiO2	interference thin film	3 °C	10%	NG	China
Joshi and Dhoble (Joshi and Dhoble, 2017)	2017	PVT	Polycrystalline	Tap water, coconut oil silicone oil	spectrum filters	Max: 33 °C	At 12.53%	NG	Nagpur, India

Table 18
Highlights of photovoltaic system micro channel cooling studies.

Research	Year	System	PV cells	Working fluid	Method	Temperature decrease	Electrical efficiency	Climate/ weather	Location
Radwan, Ookawara (Radwan et al., 2019)	2019	CPV	NG	Ethanol and Novec-7000	monolithic double-layer microchannel	27–30 °C	NG	indoor	Sohag, Egypt
Al Siyabi, Khanna (Al Siyabi et al., 2018)	2018	CPV	NG	different heat transfer fluid (HTF)	multi-layered microchannel	14.93 °C	NG	NG	NG
Radwan, Ookawara (Radwan et al., 2018b)	2018	CPVT	NG	Ethanol and acetone	monolithic double-layer microchannel	22.4 °C	NG	NG	Tokyo, Japan

on their results, increasing the number of layers not only decreases the surface of the PV panels better, but also provides a more uniform cooling performance. The small PV cell was used in the experimental setup, therefore results regarding performance response should be carefully taken into account. Radwan, Ookawara (Radwan et al., 2019) examined two different configuration of microchannel heat sinks, namely counter flow and parallel configurations. Compared to counter flow configuration, parallel flow configuration achieved effective cooling only at higher flow rates. The proposed experimental setup is quite complex, and it was obtained in laboratory circumstances.

3.2.2.2. Thermoelectric cooling. Using the thermoelectric effect, thermoelectric generators (TEGs) convert thermal energy into electricity, similar to a heat engine based on the Seebeck effect, and a thermoelectric cooler (TECs), based on the Peltier effect, provides temperature difference between the hot and cool sides of the system (Gharzi et al., 2020; Dwivedi et al., 2020; Sahin et al., 2020). In an electrical junction, the Peltier effect occurs when the heat flows in one direction while cooling occurs in the other direction. The intensity of the voltage/-current and the temperature difference determine its intensity (Kalaiselvan et al., 2018). Nevertheless, integrating PV systems with TE modules leads to some trade-offs and complex circumstances that necessitates accurate optimization. The higher temperature difference enhances the cooling performance of the PV-TEG while on the other hand, increased temperature reduces the energy conversion efficiency and lifespan of PV cells (Sahin et al., 2020; Shittu et al., 2019b). For instance, Li, Shittu (Li et al., 2019), in their experimental research on PV-TE system, recorded 10 °C raise in temperature, despite improved electrical efficiency comparing with the stand alone PV panel. Therefore, economic and optimization feasibilities and thermal matching are the essential factors to be considered in this technique since Peltier elements are not cheap. Moreover, mentioned efficiency improvement is limited to some ranges of temperature, which depends on the type of the PV installation (Enescu and Spertino, 2017). It is particularly effective in hot climates due to the high ambient temperature, and in conditions where cooling of the system is vital but convection is hard (Kane et al., 2017).

As Moh'd A and Mugdadi (Moh and Mugdadi, 2020) have indicated in their study the impacts of the ambient temperature, the wind speed, solar radiation intensity and coolant outlet temperature on the performance of the concentrated photovoltaic/thermal (CPVT) system equipped with thermoelectric cooling. The solar radiation and PVT outlet temperature were the most effective criteria in determining cooling capacity, in contrast to the ambient temperature and wind speed with almost no significant impact which was also concluded in Wu, Zhang (Wu et al., 2018b) study. Highlights of the reviewed studies are presented in Table 19. When analysing the obtained research stuides with TEG cooling approaches it was found that economic evaluation was not well discussed. Moreover there is necessity for discussion of limitations of TEG-PV cooling when considering specific climate conditions.

Table 19
Highlights of photovoltaic system thermoelectric cooling studies.

Research	year	System	PV cells	Method	Temperature decrease	Electrical efficiency gain	Climate	Location
Metwally, Mahmoud (Metwally et al., 2021)	2021	PV-TE	NG	Thermoelectric (Edenhofer et al., 2011) active cooling	<36 °C	20%	_	_
Li, Shittu (Li et al., 2019)	2019	PV-TE	crystalline silicon	Thermoelectric cooling	10 °C rise	at more than 14.0%	NG	Hefei, China
Wu, Zhang (Wu et al., 2018b)	2018	PVTCS	NG	photovoltaic-thermoelectric cooling hybrid system (PV-TCS)	About 10 °C	NG	_	China
Kane, Verma (Kane et al., 2017)	2017	PV-TE	NG	Thermoelectric module (Rajaee et al., 2020)	6–26%	1–18%	NG	NG

3.3. Combined cooling

In some cases, combining two or several cooling techniques (hybrid cooling approach) could lead to thermal utilization improvement, enhanced efficiency, and PV system's lifespan. Passive or active techniques could be combined in order to compensate for some of each drawback, namely, thermal recovery capability, the performance of cooling, cooling uniformity,etc. (Gharzi et al., 2020). Mentioned combination should be precisely accurate, as in some methods, the extracted heat is used for other purposes like hot water preparation and in some others, it is dispensed to the surroundings (Gharzi et al., 2020). Joint cooling methods are in form of passive-active techniques, among which using PCMs as a conductive cooling technique seems to be more frequent in combination with other technologies.

Table 20 contains research outputs of some combined cooling studies including both passive techniques, which are more investigated. Rajvikram, Leoponraj (Rajvikram et al., 2019) experimentally studied the cooling capacity of PCM entrenched with aluminum as a thermal conductivity enhancer and according to their investigations, the aluminum sheet accelerated the dispersion of the heat to the ambient environment while the PCM reduced the surface temperature of the PV panel. The experimental setup was quite simple one and economic evaluation was not provided. Sudhakar, Santosh (Sudhakar et al., 2021) also recorded augmentation in the PV performance, incorporating water with PCM cooling, examining the water direction, flow rate and duration. They concluded top to bottom, continuous water flow with OM 35 PCM at the Tedlar® surface of the PV panel has the best performance. Again, an economic aspect was not considered and which is critical to determine. Wongwuttanasatian, Sarikarin (Wongwuttanasatian et al., 2020) evaluated combined palm wax PCM with three different PCM heat sink containers, including grooved, tubed and finned containers, which recorded the best result. This study considered different geometries of heat sinks and that were not discussed from economic point of the view, moreover investigation was obtained on the small PV panel (20 W). They also indicated that enhancement in PV performance using PCM material is highly dependent on the solar irradiance and might not always lead to improvements.

3.3.1. Passive-active

Highlights of some of combined passive-active cooling studies are shown in Table 21 which are further evaluated in the following. Salem, Elsayed (Salem et al., 2019) experimentally investigated the combination of water and/or Al2O3/PCM mixture with variant nanoparticles mass concentrations and occupation ratio in the channels. According to their results, although Al2O3 ($\varphi = 1\%$)/PCM mixture ($\lambda_{PCM} = 25\%$) + 75% water (5.31nullkg/s.m^2) recorded the highest efficiency, the Al2O3/PCM mixture of $\lambda_{PCM} = 100\%$ recorded the best result as it provides the chance to use the cooling water. In another experimental study, Rajaee, Rad (Rajaee et al., 2020) indicated 4.52% improvement in the overall electrical efficiency of a PVTEG by utilizing both PCM and 1% nanofluid. Likewise, Siahkamari, Rahimi (Siahkamari et al., 2019) used copper microchannel tubes containing cold water in a chamber at the PV panel's backside to postpone the PCM material's melting. In all mentioned studies with nanofluids, an economic and environmental analysis was not delivered, and which are important in order to understand suitability of the cooling approaches.

Through a numerical study, Radwan, Ahmed (Radwan et al., 2016) evaluated the cooling performance of microchannel heat sink with two types of nanofluids namely, Aluminum Oxide (Al2O3)–water and Silicon Carbide (Grubišić-Čabo et al., 2016). The mentioned study (Siahkamari et al., 2019) was obtained on numerical way and compared with some available experimental data. However, the real field experiment would be beneficial. Moreover, different nanofluids were considered but selection basis is not clear. According to their investigations, at higher concentration ratio and lower Reynolds number (Re), nanofluids outperformed in enhancing the electrical efficiency of the solar cells comparing to water and could maintain the panel surface temperature at about 38 °C, while enhancing the electrical efficiency up to 19%.

The combined cooling approaches needs to be further investigated, especially on experimental basis in order to understand suitable cooling configurations for specific climates. There is also reasonable possibility to use smart and IoT based technologies in order to secure effective operation, i.e., balancing between passive and active cooling, considering specific working circumstances.

4. Discussion

In this section we have provided a holistic summary of the reviewed methods, to indicate the advantages, disadvantages, range of temperature decrease, electrical efficiency gained, and the suitable climate

Table 20
Highlights of the photovoltaic system combined passive cooling studies.

Research	year	System	PV cells	Methods	Temperature decrease	Electrical efficiency gain	Climate	Location
Sudhakar, Santosh (Sudhakar et al., 2021)	2021	PV	Polycrystalline	PCM and natural water-cooling system	5.4 °C	12.4%	NG	Chennai, India.
Wongwuttanasatian, Sarikarin (Wongwuttanasatian et al., 2020)	2020	PV	Polycrystalline	Palm wax PCM in a finned container heat sink	6.1 °C	5.3%	warm	Khon Kaen, Thailand
Rajvikram, Leoponraj (Rajvikram et al., 2019)	2019	PV	Crystalline	PCM and aluminium thermal conductivity enhancers (TCE)	10.35 °C	2%	NG	Chennai, India
Anand, Senthil Kumar (Anand et al., 2019)	2019	CPV	Polycrystalline	heat sink and loop heat pipes (LHPs)	34.8 °C	10%	NG	NG

Table 21
Highlights of the photovoltaic system combined passive-active cooling studies.

Research	year	System	PV cells	Methods	Temperature decrease	Electrical efficiency gain	Climate	Location
Rajaee, Rad (Rajaee et al., 2020)	2020	PVTEG	polycrystalline	Co3O4/water nanofluid and improved PCM (paraffin wax/ Alumina powder) heat sink	NG	+12.28%	Dry	Tehran, Iran
Awad, Radwan (Awad et al., 2020)	2020	CPV	Polycrystalline	jet impingement mini- and-microchannel heat sink with a heat spreader	21.7 °C	At 14.3%	_	_
Siahkamari, Rahimi (Siahkamari et al., 2019)	2019	PV	Monocrystalline	PCM filled with (sheep fat and paraffin wax) copper microchannel tubes	20.3 °C	24.6%–26.2% depending on water flow rate	NG	Kermanshah, Iran
Salem, Elsayed (Salem et al., 2019)	2019	PV	polycrystalline	Nano fluid and Al2O3/PCM mixture in channels	7.4–14.5 °C depends on the PCM ratio	At 10.3–13% depending on the PCM ratio	NG	Cairo, Egypt
Radwan, Ahmed (Radwan et al., 2016)	2016	CPVT	Polycrystalline	microchannel heat sink with nanofluids	At 38 °C	At 19%	_	_

condition, as shown in Table.22. This is beneficial in respect to finding suitable options to be adopted in certain circumstances.

Water-based active and passive cooling methods could have multiple advantages namely being low cost and environmentally friendly while cleaning the surface of the PV panel at the same time. Using water as a coolant is considerably more widely used as the water output could be used for thermal purposes too, making the PVT system more efficient. Depending on the method and configuration uniform and steady cooling is highly promised, especially in hot and dry climate. It worth mentioning that some water-based installations are rather complex and expensive with high maintenance expenses, making them not an ideal option for large scale PV systems. On the other hand, using Nano-fluids as a coolant is tricky as it is high cost and more energy demanding, making the cooling system more complex and with necessary consideration of the safety aspect. Regarding these issues, it is more recommended for industrial use for long-term applications. Among the fluid-based cooling approaches, liquid immersion seems to be the most controversial as despite being highly efficient in cooling terms, the durability of the system might be threatened by possible degradation in long-term. It also brings environmental concerns as well. In terms of the most electrical efficiency gained, according to the reviewed papers, spectrum filtering recorded the best results of up to 53%, followed by jet impingement and air-based active cooling by 47.67% and 44.34% respectively (however here the size of the examined PV panel/cell needs to be taken into account when discussing performance indicators). Regarding the highest temperature decrease, up to 46 °C was obtained by jet impingement cooling followed by radiative and thermoelectric cooling, both up to 36 °C.

Most of the reviewed studies were conducted experimentally which is critical in terms of verifications and reliability of the results and it is suggested for other numerical studies to couple their results with experimental supports as well. Majority of the studies were conducted on Polycrystalline PV cells technology, followed by Monocrystalline with a little difference in numbers. Hot and arid climate was the most frequent climatic and weather condition in studies, as clearly in such conditions the overheating of the PV panels is intensified. Some studies were also conducted in tropical climate, where cooling via evaporative properties is challenging due to humidity. Heat sinks and Nano-fluid based cooling techniques were tested in this climatic condition. However, relying on spectrum filtering, radiative cooling approaches are suitable for any weather conditions, regardless of the climate. That is to say, this technology has the universal potential to be applied on the commercial panels in the market. The economic assessment for the large-scale manufacturing needs to be studied though. It also worth mentioning that the majority of the reviewed papers were conducted in Asia (middle eastern and east-southern Asian countries), plus some based in Africa (Egypt) and some sparse studies in Europe and North America.

As observed through the present investigation, environmental and economic study on the feasibility of the technologies is lacked. Also, life cycle assessment (LCA) is highly recommended for future research, as only one of the reviewed studies has included.

5. Conclusions and future directions in field

Based on the herein conducted analysis of the existing research findings it is evident that cooling approaches for the photovoltaic technologies (i.e., PV panels) can ensure several important benefits, however it is also evident that they are several other key factors that are affecting cooling performance. The crucial and the most important benefit of cooling is reduction of the PV panel operating temperature and its positive impact, both on the performance improvement, as well as on durability and lifetime of the PV panels. In recent years various cooling approaches for PV panels have been investigated and that can be generally divided into active and passive cooling strategies. Beside the proper selection of the specific cooling method, different parameters will have an important role and impact on the cooling performance such as general operating circumstances (climate), the size of the PV panel, PV technology, mounting, operating configuration, etc. However, there are some key characteristics and differences between the passive and active cooling strategies in general. For instance, passive cooling would secure the lower improvement in the performance, when compared to the active cooling, and where it is usually reasonable to expect improvement in the performance ranging from about 5% up to 20% in most of the cases. In the case of the active cooling performance improvement would usually go over 10% and usually up to 30%. There are some active cooling approaches that can reach even higher performance improvement as previously mentioned, i.e., over 30% as it is for example in the case of the jet impingement or spectrum filter cooling systems. Among the reviewed studies, jet impingement resulted in the highest temperature reduction of the panel surface (over 40 °C), while spectrum filter indicated the most electrical efficiency enhancement (over 50%). When considering cooling fluids, the water or air have been used as coolants, however there are also studies with nanofluids. The water is generally better coolant than air due to thermal properties. The cooling systems with nanofluids can reach suitable performance (up to 30% mostly) as thermal properties of produced nanofluids can be suitable. However, the main issues related with use of nanofluids are related to the high unit cost, demanding preparation process (long-term stability), toxicity and finally general safety.

Beside the mentioned performance improvements, the same impact of the cooling approach can be also noted with respect to the decrease rate of the PV panel operating temperature and that is much sensible in the case of active cooling. Namely, for passive cooling it is reasonable to

Table 22
Comprehensive highlights and comparison of all studied photovoltaic system cooling methods.

Cooling technique	Method	Pros	Cons	Temperature decrease	Electrical efficiency gain	Climate/weather condition
Passive	Air-based	little volumetric change at low temperature, environment friendly, small cost, the most popular and used method in real applications, Simple with no additional material	–	Upto 20 °C	Upto 22.9%	Most weather conditions even severely cold climates
	Water-based	Using the warm water for thermal purposes, prevent dust pollution as it cleans the surface of the PV panel, environmentally friendly, simple design	Water wasting	Upto 20 °C	Upto 11.9%	Hot and dry, tropical
	Nano fluid-based	suitable and expandable in the solar industry, increase in the thermal conductivity and capacity,	makes the cooling systems more expensive and complex, despite raising the overall efficiency	Upto 2.4 °C	Upto 9.7%	tropical
	Heat pipes	high heat capacity and efficiency while being compact	costly with complicated design, not suitable for large-scale application	Upto 16.1 °C	Upto 14%	Hot and dry
	Liquid Immersion	High-efficiency improvement, and temperature reduction potential	durability of the system in the liquid, the possibilities of long-term degradation, destroying the visual parade of natural water landscapes, a threat to the aquatics life	Upto 16 °C	Upto 17.8%	Hot, cold
	Heat Sinks and Wick Structures	Simplicity, more likely to be adopted at a large scale due to their more straightforward production and lower cost	additional conductive components	Upto 30 °C	Upto 16.85%	Hot and dry, tropical
	Phase change material (PCM)	The ability to maintain considerable amount of thermal energy for temperature stabling, Reutilization of the saved heat in the back of the PV surface	less cooling effect in the peak of cool and hot months of the year, decrease in the heat absorption over time, high cost of raw materials	Upto 10.5 °C	Upto 11%	Only in hot climate with negligible temperature variations among the seasons
	Radiative	Simpler modifications with no extra additional large components, lightweight	some overestimations on its effects compared to encapsulated PV panel leading to controversial debates on its practical application and efficiency	Upto 36 °C	Upto 27%	Various conditions
Active	Air-based	Minimal materials, the most basic type of cooling, low cost	An extra amount of electricity is consumed by the pump and the fan	At 40–56 °C	Upto 44.34%	Hot and arid
	Water-based	more uniform distribution on the surface, potential of light absorption enhancement, high thermal capacity, cleanness and extended system lifespan, one of the most popular techniques, using the warm water for thermal purposes	Energy consumption, bulky. equipment degradation, high investment	At 22.3 °C	Upto 30.6%	Hot and dry
	Jet impingement	Steady and uniform cooling performance, high power	In case of mistake to calculate the number of the nuzzle, derangement might happen between the spouts of the jets	Upto 46 °C	Upto 47.67%	NG
	Spectrum filter	Lightweight, broadly installed on hybrid solar photovoltaic-thermal systems	May decrease in electricity yield	Upto 33 °C	Upto 53%	NG
	Nano-fluid base	Enhancing both the energy and exergy efficiencies, stable and suitable for long term use	agglomeration, instability, erosion, high cost, high viscosity, increase in pumping power, and lower specific heat	2 °C	Upto 24.1%	Semi-arid
	Micro channels	Simultaneous thermal and electrical energy generation improvement, more uniform cooling performance	Problems in designing for a large scale, sedimentation in case of using nanoparticles in the channel with micro-scale diameter	Upto 30 °C	NG	NG
	Thermoelectric	_	complex circumstances and trade-offs which necessitates accurate optimization, limited efficiency improvement	Upto 36 °C	Upto 20%	particularly hot climates with high ambient temperature
passive-active	thermal utilization improvement, enhanced efficiency and PV system's lifespan, compensate each other's drawbacks	Needs extremely precise design and calculations	Upto 34.8 °C	Upto 26.2%	Warm and dry	

expect an average reduction of the PV panel operating temperature usually from 10 °C to 20 °C, while in the most of the active cooling strategies, temperature decrease would range from 20 °C to about 40 °C. The very important impact on the temperature reduction rate has the size of the PV panel and with respect to that performance improvement needs to be carefully evaluated in order to secure reasonable basis for comparison. Namely, the size of the PV panel can strongly affect cooling performance, and which is crucial when comparing specific cooling strategies. Moreover, when examining several PV panels simultaneously it is mandatory to record their performance before any modification in order to determine possible differences in their referent performance in order to get more credible results with respect to the performance improvement. Due to this lack, all the existing comparisons are qualitative and, in this research, the more solid and quantitative comparison between the mutual values was provided. The important advantage of the passive cooling approaches is relatively simple design, and which is followed by reasonable overall cost. Finally, the simpler design also affects the maintenance aspect in passive cooling, and which is less demanding when compared to the active cooling. Moreover, there is no need for the operational energy in the case of the passive cooling technologies and which is beneficial regarding economic viability.

Active cooling is usually more complex in design and associated with high unit cost and where also operational energy is required to enable specific cooling. The request for operational energy is a key factor for the economic viability of the active cooling approaches, since in some cases the performance improvement can annul reached performance improvement. Because of the previous issue the most important aspect is to secure proper design and to match it for specific climate conditions. The favorable aspect of the active cooling techniques is better flexibility and possibility to secure more uniform cooling when compared to the passive cooling. The special category regarding the cooling aspect is related to the photovoltaic-thermal (PVT) systems since efficient cooling is just one goal, while other goal is to secure hot water of suitable temperature level. In mentioned circumstances there is an issue with two opposite goals, so with respect to that the key aspect is careful design, i.e., design that would match demands. Namely, the request for the higher temperature level of the hot water could potentially affect electrical efficiency of PVT systems, since an average operating temperature of the PV panel is being increasing in that case. Therefore, it is important to find proper balance between electrical and thermal performance of the PVT systems and where crucial aspect is design, as already emphasized. When analyzing the existing research efforts in this specific area it was also found that majority of the conducted studies were mainly focused on the performance analysis of different cooling strategies, while economic and especially environmental aspect were weakly investigated.

Regarding possible future research directions in the field, the following key research tasks could be summarized and that were based on the herein conducted review:

- There is obvious necessity for categorization and definition of universal evaluation framework and that should help to make proper selection of specific cooling method with respect to the climate circumstance. The general guideline that would secure better match between specific selected cooling approach and specific climate circumstances. The proposed framework should also take into account specific configuration, i.e., setup of the PV system as well as size of the PV panel,
- Each proposed PV cooling approach should be evaluated from efficiency, economic and environmental point of the view (request for integral 3 E evaluation). In that sense the future research efforts should be directed in order to enable fair comparison basis for various cooling approaches,
- There is request for necessary optimization of already examined cooling techniques in the literature in order to improve their economic viability. Optimization could lead to less use of raw resources and that would be followed by higher performance improvement,
- The combination of passive and hybrid cooling techniques (hybrid cooling) could enhance the system's cooling performance by compensating specific drawbacks between passive and cooling methods. Hybrid cooling approach can improve overall system flexibility in various operating circumstances or demanding climates. The jet impingement and spectral cooling approaches should be considered and further advanced since have the potential to reach the highest performance improvement,
- The more research efforts should be directed to the further advancements in the field of building integrated photovoltaics (BIPV) and their effective integration with suitable cooling methods for PVs, in order to create active building energy systems and by that to expand impact of PVs in building facilities,
- There is lack of experimental studies related to the cooling approach by photonic devices. Therefore, in that sense the more research efforts should be provided in order to understand potential technological limits and economic viability.

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CRediT authorship contribution statement

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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